

SE3090 ASSIGNMENT 1 | BUILDWISE PROJECT SCENARIO

Construction Materials Procurement, Delivery and Quality Management System

BuildWise - Integrated Full-Stack and Agentic AI Application

TEAM REFERENCE

This document explains the complete BuildWise scenario in one place so every team member can understand the problem, users, four components, shared workflow, four AI contributions, human approvals, web/mobile responsibilities and core data.

Real-world problem

Construction material procurement becomes difficult when material needs start at a construction site but supplier selection and purchasing decisions happen in an office. Requests, quotations, approvals, purchase orders, deliveries and inspections may be tracked through separate calls, messages, spreadsheets and paper records.

BuildWise connects those activities in one system. Site users use a Flutter mobile application for material requests, delivery receiving and quality inspection. Procurement and management staff use a React web application for supplier and quotation management, approvals, purchase orders, monitoring and administration. ASP.NET Core provides the shared secure API and business rules, PostgreSQL stores the shared relational data, and Agentic AI assists decisions while important actions remain under human control.

User roles

- 1. Site Engineer / Site Officer.** Creates material requests, tracks progress and records deliveries from the construction site.
- 2. Procurement Officer.** Maintains suppliers, records quotations, compares options and prepares procurement.
- 3. Procurement Manager.** Reviews request approvals and AI-assisted procurement recommendations, then approves, rejects or requests revision.
- 4. Quality Inspector.** Inspects delivered materials, records accepted/rejected quantities and manages non-conformance cases.
- 5. Administrator.** Manages users, roles and shared system settings.

Implementation note. User authentication and role-based authorization are shared mandatory features and should not be counted as one of the four main business components. Supplier records are business data managed by procurement staff.

Four main business components

Component	Main functionality	Business-specific operation
Material Request & Approval Management	Projects, materials, requests, request items, required dates, reasons, statuses and request approval decisions.	Submit and control a material request, preserve approval history and prevent silent changes after approval.
Supplier, Quotation & Procurement Management	Suppliers, quotation items, validity, comparison, AI-assisted recommendation, procurement approval, purchase orders and purchase-order items.	Evaluate eligible quotations and create a purchase order only after the recommendation passes validation and authorized approval.
Delivery & Material Receiving Management	Expected purchase-order information, deliveries, received quantities, damaged quantities and delivery status.	Reconcile what was ordered against what actually arrived and flag shortages, damage or partial delivery.
Quality Inspection & Non-Conformance Management	Inspections, inspection items, accepted/rejected quantities, quality decisions and non-conformance records.	Complete the inspection and create a traceable non-conformance when rejected material requires corrective action.

Each student owns one component and should implement its ASP.NET Core endpoints, PostgreSQL entities and relationships, React interfaces, Flutter interfaces, tests, documentation, Git evidence and one identifiable Agentic AI contribution.

Example complete scenario

A Site Engineer submits this objective using Flutter:

“Request 250 bags of OPC cement for the Colombo apartment project, required by 22 August for ground-floor column concreting.”

The request is reviewed and approved for procurement. The Procurement Officer then records three supplier quotations:

- **Supplier A:** active, full 250 bags available, valid quotation, total LKR 525,000.
- **Supplier B:** suspended, full 250 bags available, lower total of LKR 510,000.
- **Supplier C:** active and cheapest, but only 200 bags are quoted.

Workflow

1. Flutter submission. The Site Engineer logs in, selects the project and material, enters 250 bags, the required date and reason, and submits the request.

2. ASP.NET Core processing. The API validates the JWT, role, project, material, quantity and date, then saves the request and items in PostgreSQL.

3. Request Validation & Planning Agent. The component-owned agent checks completeness, quantities, dates and urgency, then produces a structured request summary and planning flags. It does not approve the request.

4. Request approval. An authorized manager reviews the request and records Approve, Reject or Revision Requested. Approved requests may continue to procurement.

Workflow continued

5. React procurement preparation. The Procurement Officer records supplier quotations. Each quotation is linked to the original material request, and each quotation item is linked to the corresponding request item.

6. Quotation & Supplier Analysis Agent. This component-owned agent compares eligible quotations using structured data such as supplier status, quotation validity, quantities and calculated totals. It identifies that Supplier B is suspended and Supplier C does not cover the full quantity.

7. Structured recommendation. The agent recommends Supplier A and returns a short structured rationale/evidence summary, ranked alternatives and warnings. BuildWise stores observable summaries and results - not private model chain-of-thought.

Deterministic validation

Before the recommendation can reach procurement approval, ASP.NET Core verifies fixed rules:

- the material request exists and has the required approval;
- the recommended supplier exists and has Active status;
- the recommended quotation exists and is not expired;
- quotation items belong to the requested material items;
- the recommended quantity covers the requested quantity;
- the quotation total matches the quotation-item calculation;
- the recommendation follows the required structured output schema;
- purchase-order creation is blocked until authorized procurement approval exists.

Human approval

8. Procurement Manager review. The workflow becomes **AwaitingApproval**. The Procurement Manager uses React to review the recommended supplier, structured rationale, warnings, validation results and execution summary.

The Procurement Manager can:

- **Approve**
- **Reject**
- **Request Revision**

9. Final procurement execution. If approved, ASP.NET Core records the agent workflow approval and creates the purchase order and purchase-order items from the selected quotation. Purchase orders remain owned by Component 2; Component 3 only consumes approved purchase orders for receiving.

10. Flutter status update. The Site Engineer receives the approved procurement status through the mobile application.

Therefore, the assessed procurement workflow is:

Flutter → ASP.NET Core → PostgreSQL → Agentic AI → React approval → ASP.NET Core / PostgreSQL → Flutter status update

Delivery and quality continuation

BuildWise continues the same traceable procurement record after purchase-order approval:

11. Delivery receiving. When the supplier arrives, the Site Officer opens the approved purchase order in Flutter and records the actual delivery. Example: 250 bags ordered, 240 bags received and 5 of the received bags visibly damaged.

12. Delivery Discrepancy Agent. This component-owned agent compares ordered, received and damaged quantities. It produces a structured discrepancy summary such as shortage = 10 bags and damaged = 5 bags, with warnings for partial or abnormal delivery. The backend independently validates the arithmetic before saving the result.

13. Quality inspection. The Quality Inspector opens the delivery in Flutter, performs the inspection and records accepted/rejected quantities. Example: 235 bags accepted and 5 bags rejected.

14. Quality Risk & Non-Conformance Agent. This component-owned agent analyses the inspection result and existing supplier-quality evidence, then suggests risk flags and an appropriate non-conformance/corrective-action summary when required. The Quality Inspector remains responsible for the recorded inspection decision.

15. Non-conformance. If rejected material requires corrective action, BuildWise creates a non-conformance record linked to the affected inspection item and tracks it until resolved or closed.

Complete BuildWise business flow

Material Request → Request Approval → Quotations → AI-Assisted Procurement Evaluation → Manager Approval → Purchase Order → Delivery → AI Discrepancy Analysis → Quality Inspection → AI Quality-Risk Assistance → Acceptance / Non-Conformance

Four specialized Agentic AI contributions

Owned component	Agent	Responsibility
Component 1 - Material Request & Approval	Request Validation & Planning Agent	Checks completeness, quantities, dates and urgency; creates a structured request plan/flags.
Component 2 - Supplier, Quotation & Procurement	Quotation & Supplier Analysis Agent	Compares eligible quotations and supplier evidence; returns ranked options, structured rationale and warnings.
Component 3 - Delivery & Receiving	Delivery Discrepancy Agent	Compares ordered vs received/damaged quantities and flags shortages, partial deliveries or discrepancies.
Component 4 - Quality Inspection & NCR	Quality Risk & Non-Conformance Agent	Assists with inspection-risk interpretation and prepares structured NCR/corrective-action suggestions.

Shared safety rule: deterministic backend validation is a system-wide control, not a fifth AI agent. Agents may analyse and recommend, but authorization, calculations, state transitions and other high-impact actions are enforced by ASP.NET Core and authorized users.

React web application

The React application is mainly for procurement, management, administration and monitoring:

- material-request review and approval visibility;
- supplier management and supplier status control;
- quotation entry and side-by-side comparison;
- Agentic AI workflow start, status, structured summaries and execution-history views;
- recommended supplier, ranked alternatives, structured rationale/evidence summary, warnings and validation results;
- Procurement Manager Approve / Reject / Request Revision controls;
- purchase-order creation and management;
- delivery discrepancy and quality/non-conformance history;
- user and role administration;
- search, filtering, sorting, pagination, dashboards and reports.

Flutter mobile application

The Flutter application is mainly for construction-site operations:

- registration/login/logout and secure JWT token storage;
- create material requests and select required dates;
- track request and procurement status;
- view approved purchase-order information required for site receiving;
- record delivery quantities, damaged quantities and delivery references;
- perform quality inspections and record accepted/rejected quantities;
- view non-conformance status where relevant;
- receive push notifications for important workflow changes.

Third-party integration

Firebase Cloud Messaging (FCM) can be integrated through ASP.NET Core to send mobile notifications such as request approval, procurement recommendation awaiting manager approval, purchase-order creation, delivery discrepancy and inspection completion/rejection. Firebase credentials must remain server-side and failures must be handled safely.

Core data used by BuildWise

Area	Main entities
Identity	users, roles, user_roles
Material Request	projects, materials, material_requests, material_request_items, approvals
Procurement	suppliers, quotations, quotation_items, purchase_orders, purchase_order_items
Delivery	deliveries, delivery_items
Quality	inspections, inspection_items, non_conformances
Agentic AI	agent_workflows, agent_workflow_steps, agent_approvals